

Spring Semester 2025–2026

Course Name: Design of Machine Elements

Team: Team 1



Design, Fabrication and Analysis of a Robotic Lawn Mower

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Abstract

This report presents the structural and mechanical design of a lawn mower robot developed for grass-cutting applications on a predefined path. The robot is designed using a compact wheeled locomotion mechanism powered by DC motors and controlled using a Raspberry Pi-based control system. Unlike tracked locomotion systems, the present design uses conventional wheels for mobility, resulting in reduced complexity, lower power consumption, simpler manufacturing, and improved maneuverability.

The robot chassis was designed using Fusion360 software and manufactured using 3D printing techniques. The structure supports all major components including the wheel assemblies, motor mounts, cutting mechanism, battery system, controller unit, and sensor interfaces.

The report includes detailed mechanical analysis, stress calculations, deformation analysis, factor of safety evaluation, and finite element analysis (FEA). Structural stresses developed due to motor loading, blade vibration, and operational forces were studied using analytical and simulation methods.

Special attention was given to stress concentration effects around mounting holes, and geometric discontinuities. Equivalent stress distribution, total deformation, and safety factor analysis were carried out using ANSYS Static Structural analysis.

The obtained results confirm that the designed chassis structure possesses sufficient stiffness and strength to safely withstand operational loading conditions. The proposed lawn mower robot provides a compact, efficient, and scalable solution for modern lawn maintenance systems.

1 Introduction

Robotic systems have become increasingly important in modern engineering and automation applications. Robotic lawn mowers are among the most practical domestic automation systems developed to reduce manual labor and improve operational efficiency in lawn maintenance.

Conventional lawn mowing systems require continuous human effort and supervision. These systems may also involve fuel consumption, operational noise, and maintenance complexity. Lawn mower robots overcome these limitations by enabling intelligent and grass-cutting operations with minimal human intervention.

The lawn mower robot developed in this project uses a four-wheel locomotion mechanism powered by DC motors. The structure is designed to carry the controller system, power supply, cutting blade

assembly, and supporting electronics while maintaining sufficient structural rigidity.

The robot chassis was designed using CAD modeling software and manufactured using 3D printing technology. Structural analysis was performed to ensure that the chassis remains safe under operational loading conditions.

1.2 Objectives

1. To design a structurally safe lawn mower robot.
2. To develop a compact wheeled locomotion mechanism.
3. To perform stress and deformation analysis of the chassis.
4. To evaluate factor of safety under operational loading.
5. To perform finite element analysis using ANSYS.
6. To study stress concentration effects near mounting interfaces.

2 Design Overview

The lawn mower robot consists of several integrated subsystems including:

- Main chassis
- Wheel drive system
- DC geared motors
- Rotary cutting blade
- Raspberry Pi controller
- Motor drivers
- Battery system

The robot uses conventional wheels for locomotion instead of tracked wheels. This design choice reduces mechanical complexity and improves energy efficiency.

2.1 CAD Model of Chassis

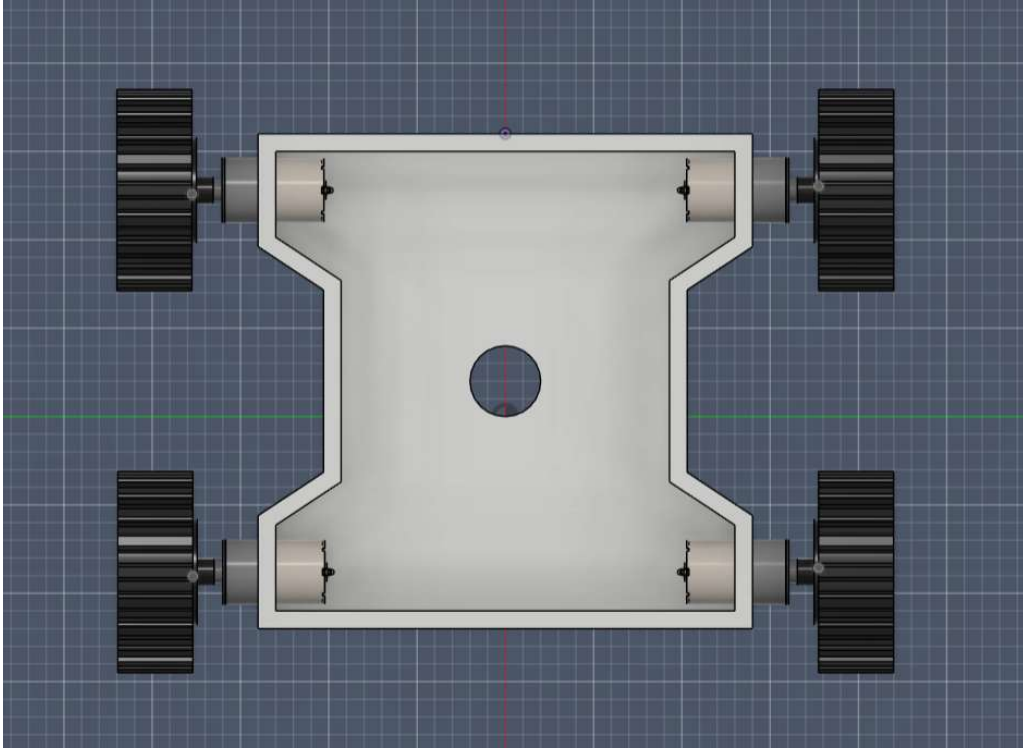


Figure 1 – Top profile of lawn mower robot

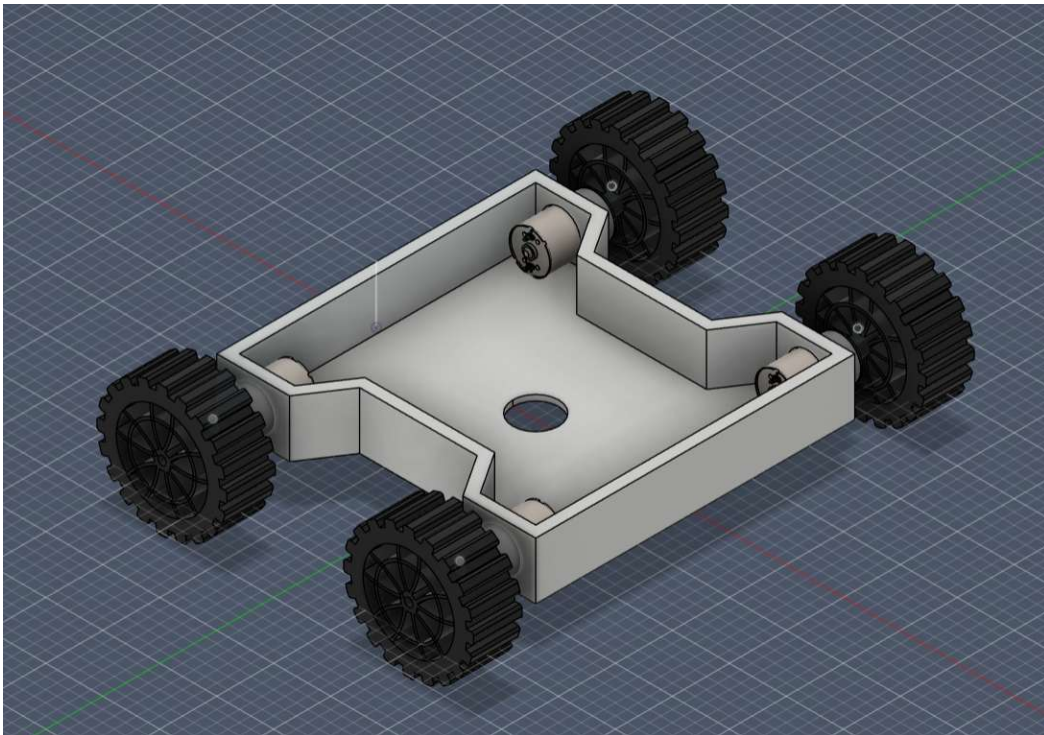


Figure 2 – Top side profile of lawn mower robot

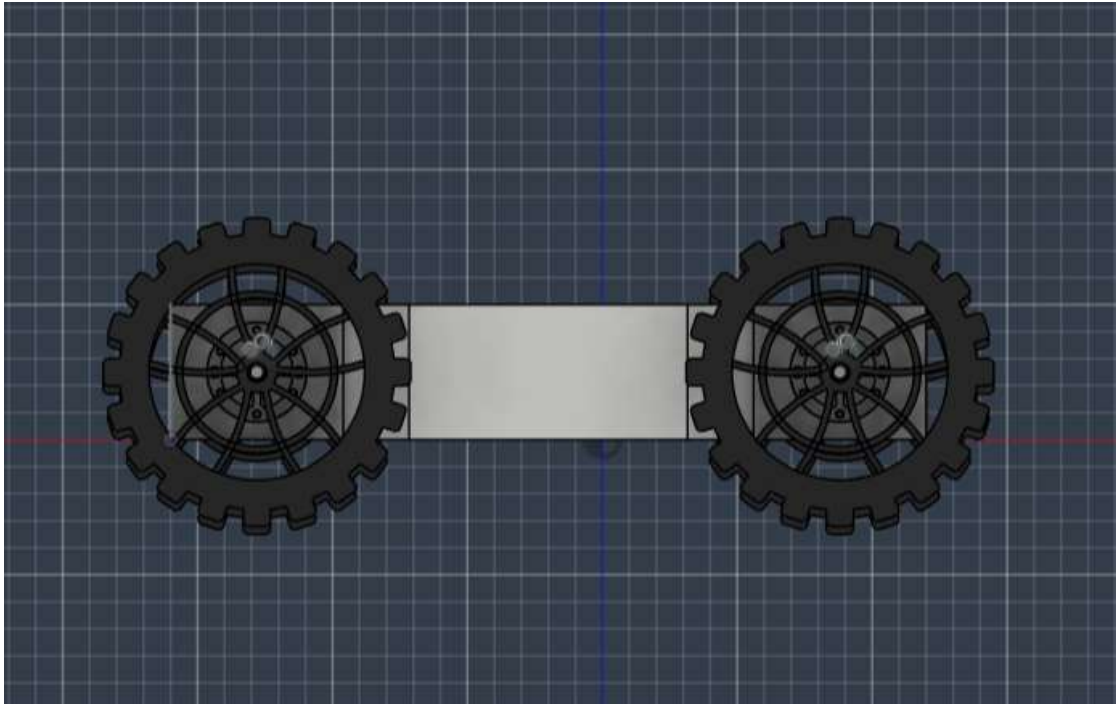


Figure 3 – Side profile of lawn mower robot

3 Working Principle

The lawn mower robot operates using independently driven conventional wheels. Four DC motors are connected to the wheels to provide motion and steering control.

The robot uses differential wheel speed control for turning:

- Equal wheel speed → Straight motion
- Higher speed on left wheels → Right turn
- Higher speed on right wheels → Left turn
- Opposite wheel rotation → Zero-radius turning

A centrally mounted cutting blade rotates at high speed to perform grass-cutting operations.

4 Mechanical Components

The major mechanical components used in the robot are:

- Chassis frame
- Wheel assemblies
- DC geared motors
- Blade support structure

- Cutting blade
- L Brackets

The chassis structure is manufactured using 3D printed PLA material due to its lightweight nature and ease of manufacturing.

5 Force Analysis

The robot experiences multiple forces during operation including:

- Weight of chassis
- Motor reaction forces
- Ground reaction forces
- Blade vibration forces
- Wheel friction forces

The force analysis is performed using free body diagrams and equilibrium equations.

$$\Sigma F_x = 0$$

$$\Sigma F_y = 0$$

$$\Sigma M = 0$$

These equations are used to determine reaction forces and internal loading within the chassis structure.

6 Stress Calculation

Stress calculation was carried out to evaluate the structural safety of the chassis.

The major stresses considered include:

- Torsional stress
- Bending stress
- Shear stress
- Principal stress

The stress equations used are:

$$\sigma = F/A$$

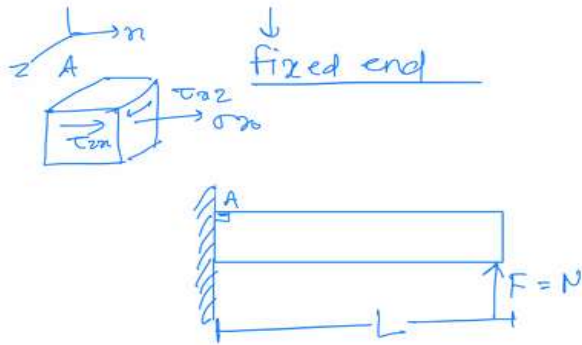
$$\sigma_b = Mc/I$$

$$\tau = V/A$$

The Von-Mises was used to determine critical loading conditions.

PLA mass $\Rightarrow 0.2 \text{ kg}$
 Motor mass $\Rightarrow 0.1 \times 4 \Rightarrow 0.4 \text{ kg}$
 L-bracket $\Rightarrow 0.03 \times 4 \Rightarrow 0.12 \text{ kg}$
 RPi 4 $\Rightarrow 0.3 \text{ kg}$
 L298N $\Rightarrow 0.01 \times 2 \Rightarrow 0.02 \text{ kg}$
 battery $\Rightarrow 0.6 \text{ kg}$
 cutting motor $\Rightarrow 0.25 \text{ kg}$
 Total weight $\Rightarrow 1.89 \text{ kg} = 1.9 \text{ kg}$
 Total force $\Rightarrow 1.9 \times 9.81$
 $\Rightarrow 18.63 \text{ N}$

Assuming total force acting at the centroid of the chassis for the simplification.



$$F = 14.65 \text{ Newton}$$

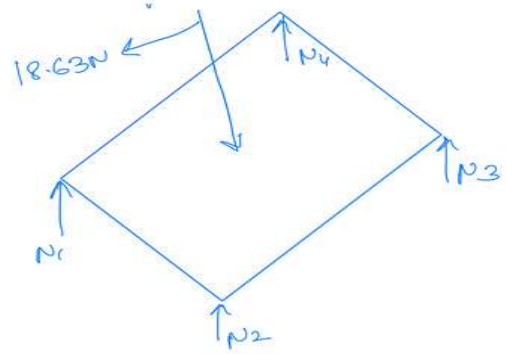
$$L \Rightarrow 0.05 \text{ meter}$$

The motor delivers driving Torque T through the shaft to spin the wheel, so it's actually combination of bending and torsion.

Therefore,

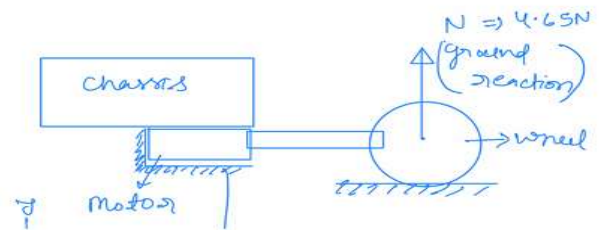
Bending moment at the fixed end.

$$M \Rightarrow F \times L$$



$$N = N_1 = N_2 = N_3 = N_4 \Rightarrow \frac{18.63}{4}$$

$$\Rightarrow 4.65 \text{ N}$$



$$\Rightarrow N \times L$$

$$\Rightarrow 4.65 \times 0.05$$

$$M \Rightarrow 0.232 \text{ Nm}$$

Bending stress

$$\sigma_b = \frac{Mc}{I}$$

$$I = \frac{\pi d^4}{64}$$

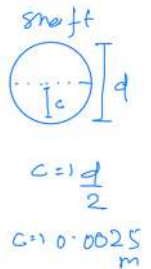
$$\Rightarrow \frac{3.14 \times (0.005)^4}{64}$$

$$I \Rightarrow 3.06 \times 10^{-11}$$

$$\sigma_b = \frac{0.232 \times 0.0025}{3.06 \times 10^{-11}}$$

$$\sigma_b \Rightarrow 1.89 \times 10^7$$

$$\sigma_b \Rightarrow 18.95 \text{ MPa}$$



Torsional shear

$$\tau = \frac{T_x}{J}$$

Motor's Stall Torque $\Rightarrow 1.117 \text{ Nm}$

$$\tau = \frac{1.11 \times 0.0025}{\frac{\pi d^4}{32}}$$

$$\tau = \frac{32 \times 1.11 \times 0.0025}{3.14 \times (0.005)^4}$$

$$\tau = 45.22 \text{ MPa}$$

now

$$\tau_{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau^2}$$

$$\tau_{max} = \sqrt{\left(\frac{18.95 - 0}{2}\right)^2 + (45.22)^2}$$

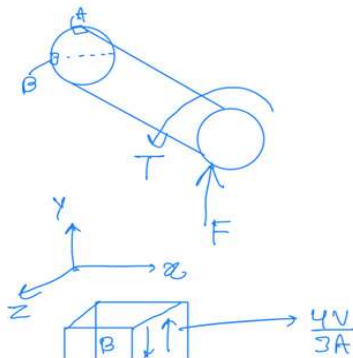
$$\sigma_{vm} = 80.57 \text{ MPa}$$

Factor of safety

$$N = \frac{S_y}{\sigma_{vm}}$$

$$N = \frac{490}{80.57} \Rightarrow 6.08$$

For B point.



$$\tau_{max} = \sqrt{89.77^2 + 2044.84}$$

$$\tau_{max} = 46.20 \text{ MPa}$$

Principle stresses

$$\sigma_{1,3} = \left(\frac{\sigma_x + \sigma_y}{2}\right) \pm \tau_{max}$$

$$\sigma_{1,3} = 9.475 \pm 46.20$$

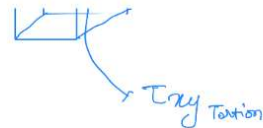
$$\sigma_1 = 55.675 \text{ MPa}$$

$$\sigma_3 = -36.725 \text{ MPa}$$

Von-mises Stresses

$$\sigma_{vm} = \sqrt{\sigma_1^2 + \sigma_3^2 - \sigma_1 \sigma_3}$$

$$\sigma_{vm} = \sqrt{3099.14 + 1348.72 + 2044.66}$$



$$\tau_{total} = \tau_{xy} + \tau_{bending}$$

$$\Rightarrow 45.22 + \frac{4 \times 4.65}{3 \times \pi d^2}$$

$$\Rightarrow 45.22 + \frac{4 \times 4 \times 4.65}{3 \times 3.14 \times (0.005)^2}$$

$$\Rightarrow (45.22 + 0.315) \text{ MPa}$$

$$\tau_{total} = 45.53 \text{ MPa}$$

Principle stresses

$$\sigma_{1,3} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{total}^2}$$

$$\Rightarrow 0 \pm \sqrt{0 + 2073.43}$$

$$\sigma_3 = \pm 45.53 \text{ MPa}$$

Von mises Stresses

$$\sigma_m = \sqrt{\sigma_1^2 + \sigma_3^2} = \sigma_1$$

$$\sigma_m = 78.86 \text{ MPa}$$

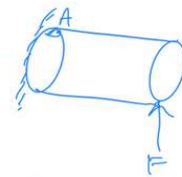
Factor of Safety

$$N = \frac{S_y}{\sigma_m}$$

$$N = \frac{490}{78.86} = 6.21$$

⇒ shaft is under fully reversed bending and torsion.

Material of shaft is mild steel.



$$S_{ut} = 370 - 620 \text{ MPa}$$

$$F = N = 4.65 \text{ N}$$

$$M = F \times l = 4.65 \times 0.05$$

$$= 0.232 \text{ Nm}$$

$$C = \frac{d}{2} = \frac{0.005}{2} = 0.0025 \text{ m}$$

$$(\sigma_a)_{\text{nom}} = \frac{Mc}{I} = \frac{0.232 \times 0.0025}{\frac{\pi d^4}{64}} = 18.95 \text{ MPa}$$

$$(\tau_a)_{\text{nom}} = \frac{Tr}{J} = \frac{1.11 \times 0.0025}{\frac{\pi d^4}{32}} = 45.22 \text{ MPa}$$

$K_T = 2$; For simplistic design
 $r = \infty$

Now, Notch sensitivity

$$q = \frac{1}{1 + \sqrt{\frac{a}{r}}} = 1$$

$$S_0 = K_f = 1 + q(K_t - 1)$$

$$K_f = 2$$

$$\sigma_a = K_f (\sigma_a)_{\text{nom}} = 2 \times 18.95 = 37.9 \text{ MPa}$$

$$\tau_a = K_f (\tau_a)_{\text{nom}} = 2 \times 45.22 = 90.44 \text{ MPa}$$

Now, Principle stresses are,

$$\sigma_{1,3} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau^2}$$

$$\begin{aligned} \sigma_a &= \sigma_x, \sigma_y = 0, \tau = \tau_a \\ \Rightarrow \sigma_{1,3} &= \frac{\sigma_a}{2} \pm \sqrt{\left(\frac{\sigma_a}{2}\right)^2 + \tau_a^2} \\ &= \frac{37.9}{2} \pm \sqrt{\left(\frac{37.9}{2}\right)^2 + (90.44)^2} \end{aligned}$$

$$= 18.95 \pm \sqrt{(18.95)^2 + (90.44)^2}$$

$$= 18.95 \pm 92.40$$

$$\sigma_1 = 111.35 \text{ MPa} \quad \sigma_3 = -73.45 \text{ MPa}$$

Now, Von-mises stresses,

$$\begin{aligned} \sigma' &= \sqrt{\sigma_1^2 + \sigma_3^2 - \sigma_1 \sigma_3} \\ &= \sqrt{17793.72 + 8178.65} \end{aligned}$$

$$\sigma' = 161.5 \text{ MPa}$$

$$S_{ut} = 450 \text{ MPa}$$

Now,

$$S_e' = 0.5 S_{ut}$$

$$S_e' = 0.5 \times 450$$

$$S_e' = 225 \text{ MPa}$$

Now, Corrected S_e ,

$$S_e = C_{load} \cdot C_{surf} \cdot C_{temp} \cdot C_{reliability} \cdot C_{size} \times S_e'$$

For bending,

$$C_{\text{bend}} = 1$$

For C_{size} ,

$$\text{degv} = \sqrt{\frac{A_{95}}{0.766}} = d = 0.005 \text{ m}$$

For $\text{degv} < 8 \text{ mm}$,

$$\therefore C_{\text{size}} = 1$$

For C_{Temp} at room is 1.

$$C_{\text{Temp}} = 1$$

Assuming 99.99% reliability,

$$C_{\text{reliability}} = 0.702$$

For $C_{\text{surface}} \neq A (\text{Sub})^b$

For grounded

$$A = 1.58$$

$$b = -0.085$$

$$\therefore C_{\text{surface}} = 1.58 (0.25)^{-0.085}$$

$$= 0.997$$

Now, Corrected S_e ,

$$S_e = C_{\text{bend}} \cdot C_{\text{surf}} \cdot C_{\text{reli}} \cdot C_{\text{Temp}} \cdot C_{\text{size}} \cdot S_e'$$

$$= 1 \times 0.997 \times 0.702 \times 1 \times 1 \times 225$$

$$S_e = 157.47 \text{ MPa}$$

For Factor of safety,

$$\text{FOS} = \frac{S_e}{\sigma_1}$$

$$= \frac{157.47}{161.15}$$

$$\text{FOS} = 0.97$$

7 Finite Element Analysis (FEA)

Finite Element Analysis was performed using ANSYS Static Structural analysis to evaluate:

- Equivalent (von-Mises) stress distribution
- Total deformation
- Safety factor
- Stress concentration regions

7.1 Equivalent Stress Analysis

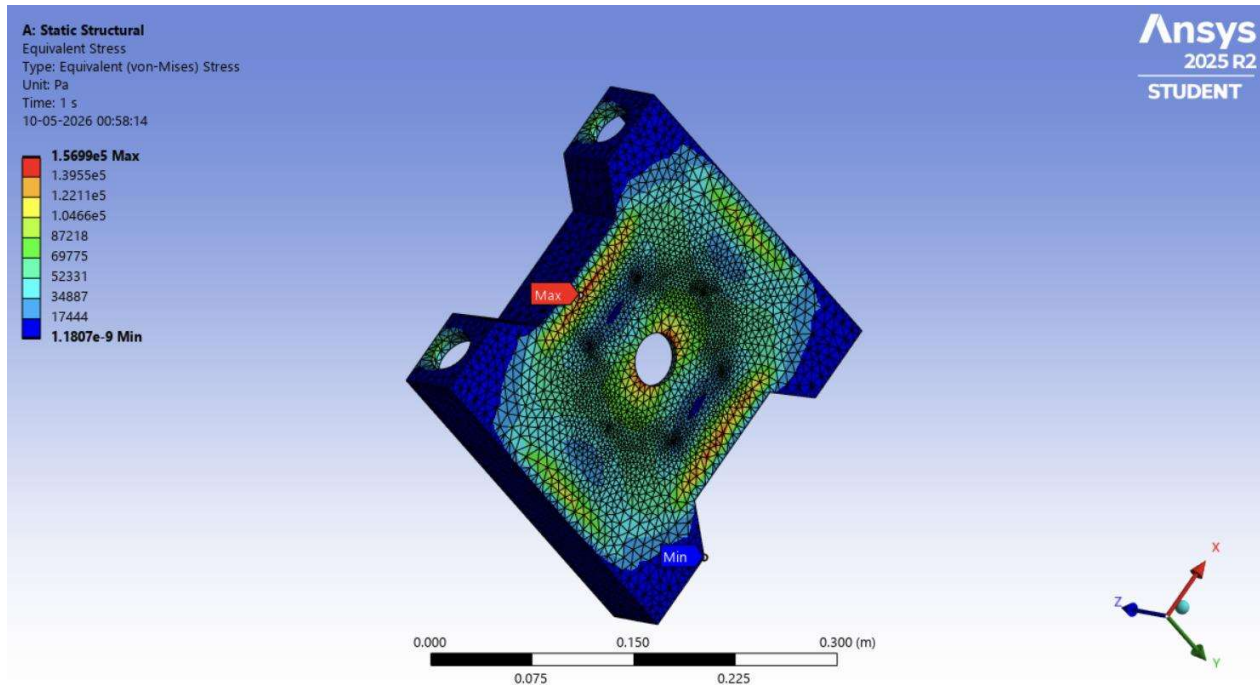


Figure 4 – Equivalent (von-Mises) stress distribution in the chassis

The equivalent stress analysis shows that the maximum stress occurs near the central mounting hole due to stress concentration effects. The peak stress value obtained from ANSYS is approximately 0.156 MPa, which is significantly lower than the yield strength of PLA material.

The stress distribution indicates smooth load transfer throughout the chassis, confirming structural stability.

7.2 Total Deformation Analysis

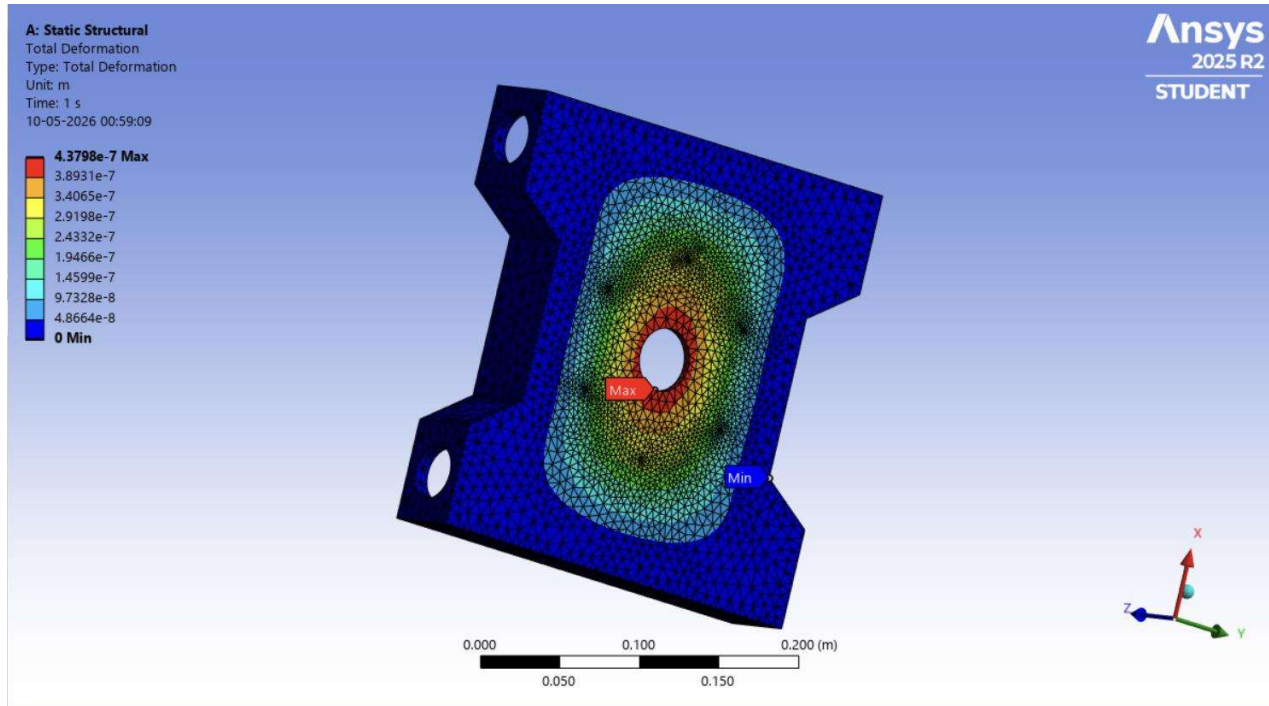


Figure 5 – Total deformation of the chassis

The deformation analysis shows that the maximum deformation occurs near the center region surrounding the mounting hole. The maximum deformation value obtained is approximately $4.37\text{e-}4$ mm.

This deformation is extremely small and confirms that the chassis possesses adequate stiffness under operational loading conditions.

7.3 Factor of Safety Analysis

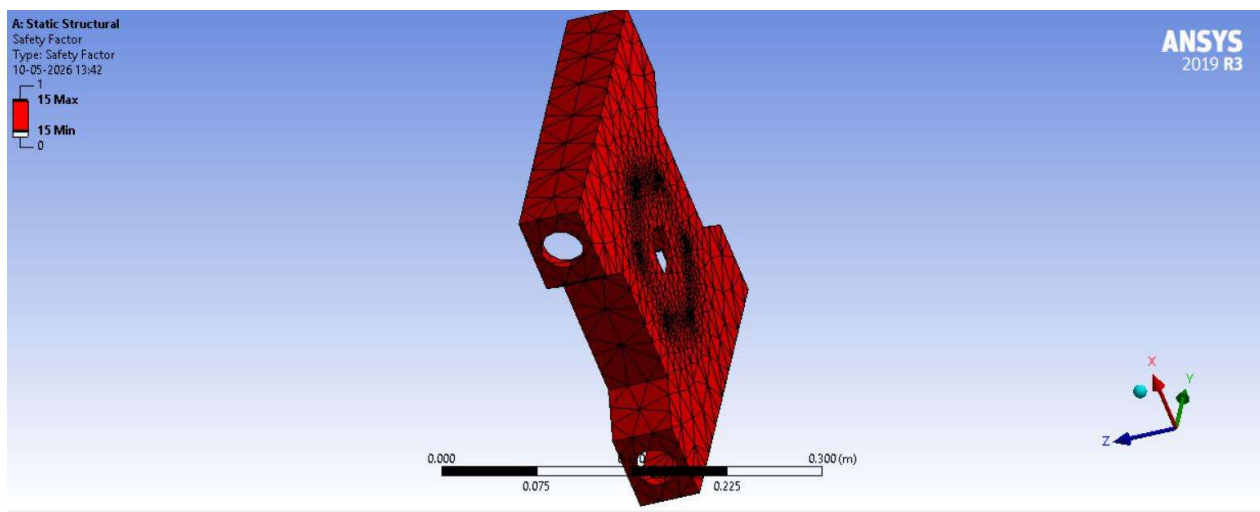


Figure 6 – Safety factor analysis of the chassis

The safety factor analysis confirms that the entire structure operates within safe loading limits. The minimum factor of safety obtained from the analysis is approximately 15, which is significantly higher than the recommended minimum design value.

This confirms that the chassis structure is mechanically safe for operational use.

8 Fatigue Analysis

Fatigue analysis was carried out on the lawn mower robot chassis to evaluate its ability to withstand repeated and fluctuating loads during continuous operation. The chassis is subjected to cyclic stresses due to wheel motion, blade rotation, vibration, and interaction with uneven ground surfaces.

The fatigue study was performed using ANSYS Static Structural and Fatigue tools. The equivalent alternating stress distribution was analyzed to identify critical regions that may experience repeated stress cycles. The maximum equivalent alternating stress obtained was approximately $1.21 \times 10^5 \text{ Pa}$, which is significantly lower than the allowable fatigue strength of the selected material.

The fatigue life result showed a minimum life of 1×10^6 cycles, indicating that the structure can sustain a very large number of loading cycles without fatigue failure under normal working conditions. The safety factor obtained from the fatigue analysis was approximately 15, which confirms that the chassis design is highly safe and reliable for long-term operation.

The maximum alternating stress was mainly concentrated near the hole regions and edge transitions due to stress concentration effects. However, the stress values remained well within safe limits.

The fatigue analysis confirms that the lawn mower robot chassis possesses excellent fatigue resistance and can operate safely under repeated loading and vibrational conditions. The obtained fatigue life and high safety factor indicate that the design is structurally durable and suitable for long-term practical applications.

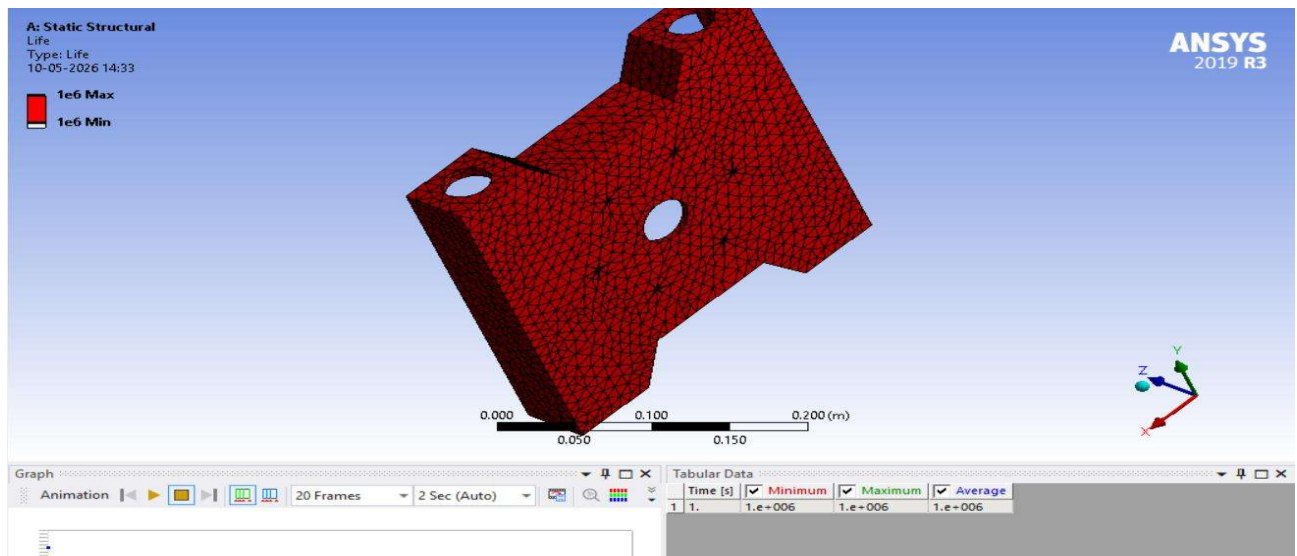


Figure 7 –Life analysis of the chassis

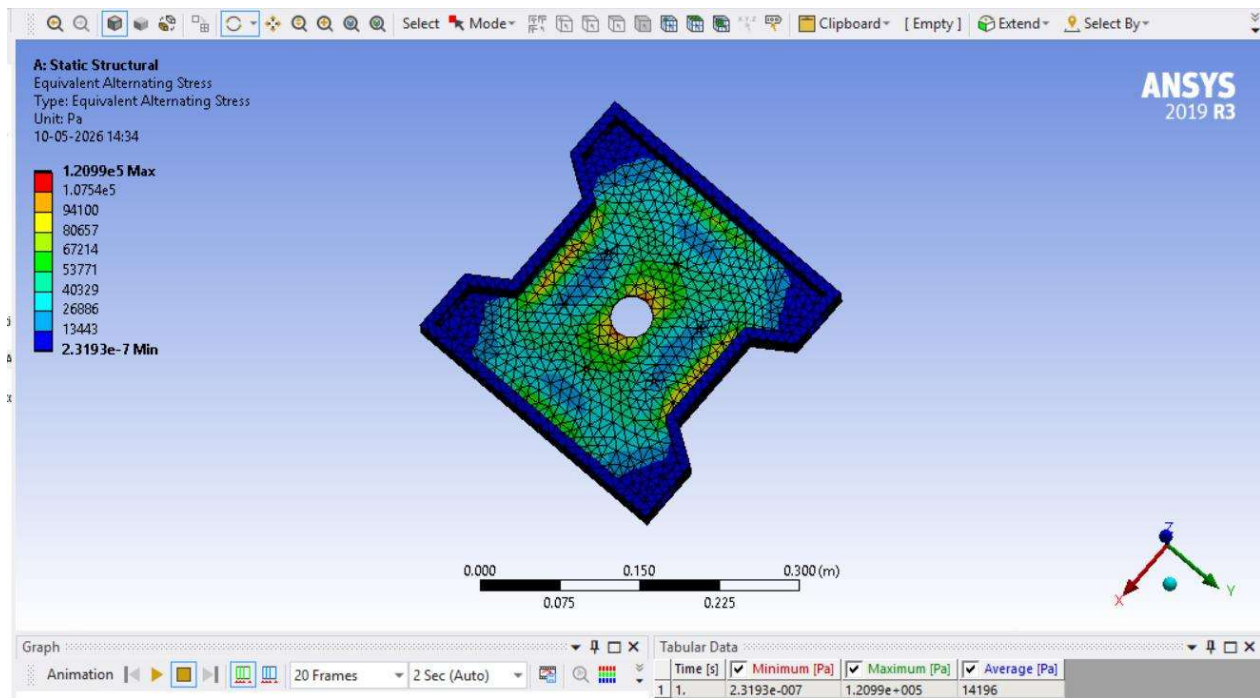


Figure 8 – Equivalent alternating stress analysis of the chassis

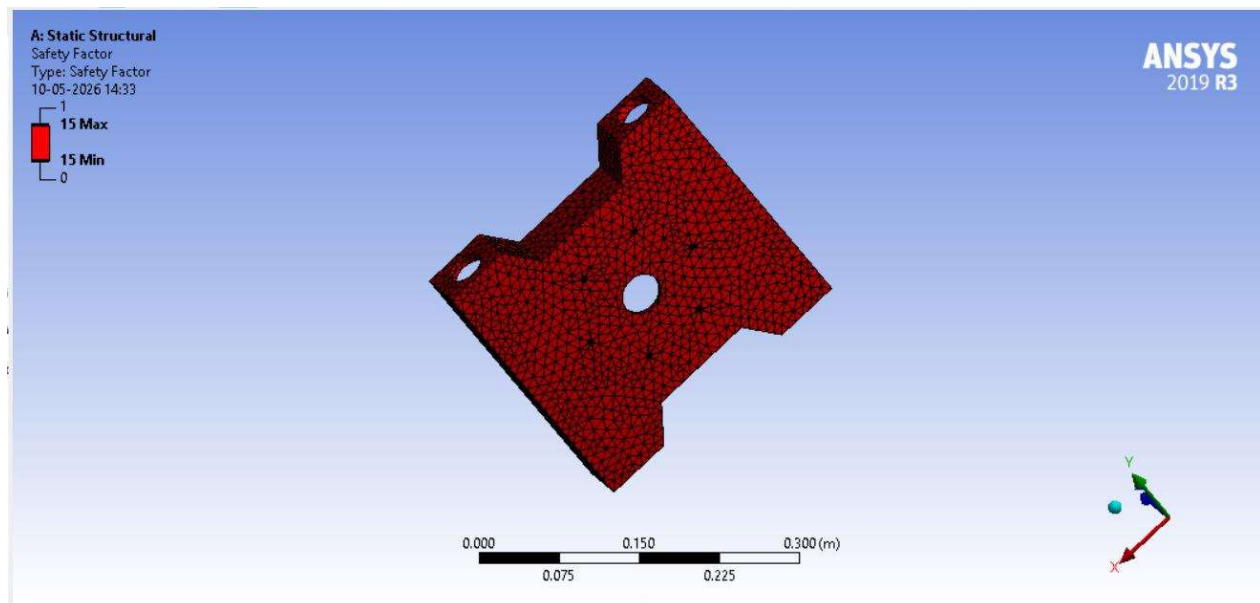


Figure 9 –FOS analysis of the chassis

9 Vibrational Analysis

Vibration analysis was carried out on the lawn mower robot to evaluate the dynamic stability of the chassis under operating conditions. The analysis was performed using ANSYS Modal and Harmonic Response tools to identify the natural frequencies of the structure and study its response under harmonic excitation.

The modal analysis results showed that the first natural frequency of the structure was approximately 319.72 Hz, while higher modes were obtained at progressively increasing frequencies up to 1553.7 Hz. The obtained modal frequencies indicate that the chassis possesses adequate stiffness and structural rigidity.

Harmonic response analysis was performed over a frequency range of 40 Hz to 400 Hz to study the variation of displacement, force amplitude, and phase angle with excitation frequency. The displacement response showed a gradual increase with frequency and reached a maximum value of displacement approximately 2.97×10^{-8} m near 320 Hz, which closely matches the first natural frequency obtained from modal analysis. Similarly, the force response exhibited a peak amplitude of approximately 1.63×10^{-2} N near the same frequency range, confirming resonance behavior.

The phase angle response showed a rapid phase shift near the resonance region, which is characteristic of vibrating mechanical systems. However, the obtained displacement and force amplitudes remained very small, indicating that the structure experiences minimal vibrational deformation under operating conditions.

The vibration analysis confirms that the lawn mower robot chassis is dynamically stable and structurally safe during operation. The observed resonance behavior near the first natural frequency does not produce excessive vibration levels, thereby ensuring reliable performance and reduced risk of structural failure during continuous working conditions.

9.1 Important Graphs

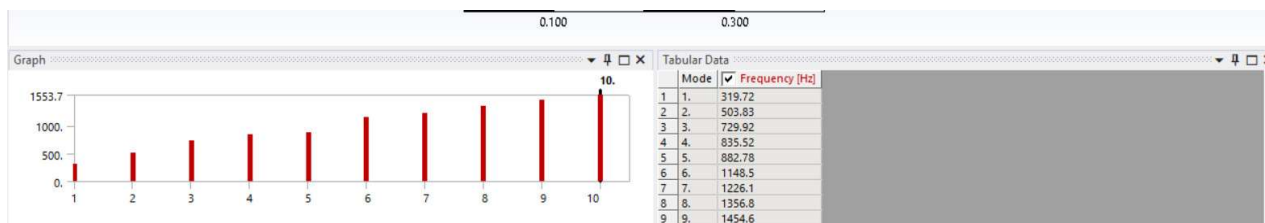


Figure 10 : Modal Analysis Frequency Plot

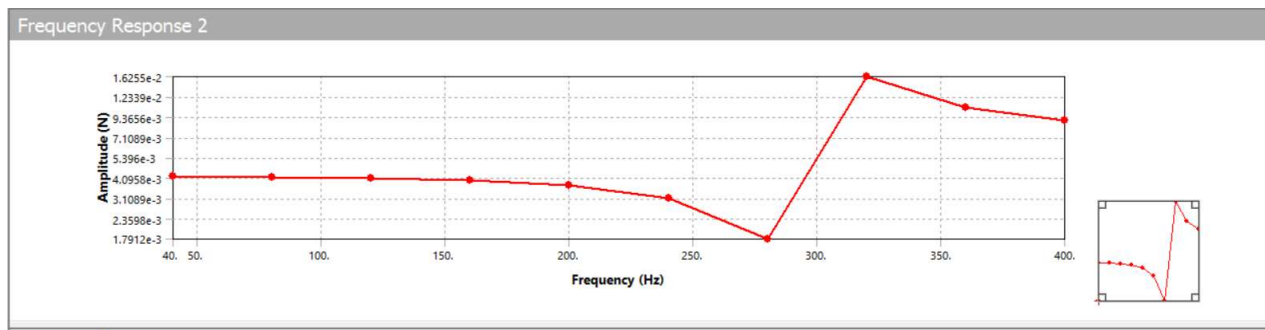


Figure 11: Harmonic Response – Force Amplitude vs Frequency

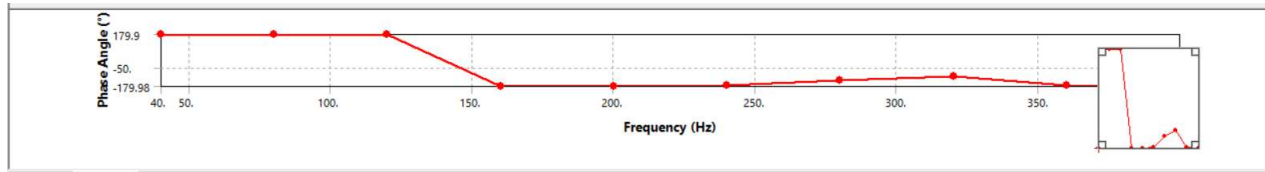


Figure 12 : Phase Angle Variation with Frequency

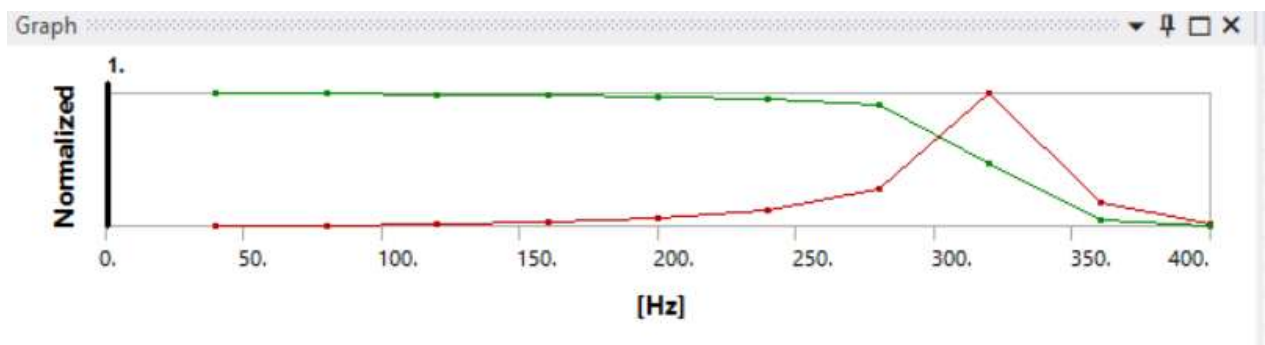


Figure 13 : Normalized Frequency Response Graph

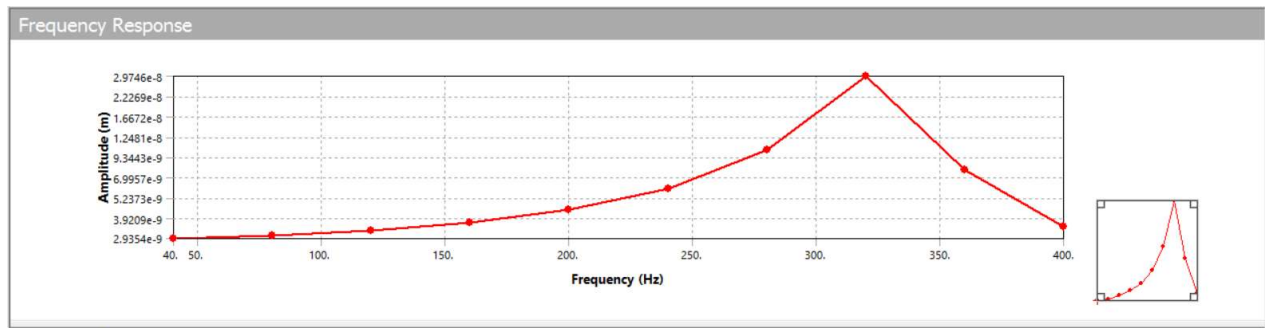


Figure 14 : Harmonic Response – Displacement vs Frequency

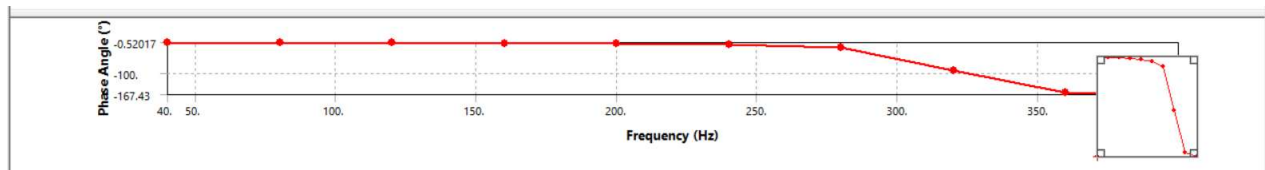


Figure 15 : Phase Angle for Displacement Response

10 Code Implementation and Control System

The lawn mower robot was programmed using the Arduino IDE and implemented on an Arduino Due microcontroller. The control algorithm was developed to achieve controlled motion of the robot along a predefined square path while simultaneously controlling the blade motor. The program included motor control, motion control, path planning, and timing-based navigation functions.

The robot utilizes three L298N motor driver modules for controlling the drive motors and auxiliary motor system. Two motor drivers were used to control the front and rear wheel motors, while the third driver controlled the auxiliary motor responsible for blade rotation. Dedicated enable and direction pins were assigned for each motor to ensure proper movement and directional control.

The navigation algorithm was based on a square path traversal of dimensions 4 ft × 4 ft. The motion sequence consisted of forward movement followed by a 90° pivot turn, repeated four times to complete the square trajectory. Timing-based calibration constants such as seconds per meter and seconds per 90° turn were used to estimate movement distances and turning angles.

Separate functions were implemented for left-side and right-side motor control, enabling accurate forward and backward wheel movement. Additional functions such as `moveForward()`, `pivotTurnRight90()`, and `driveSquarePath()` were created to simplify the motion logic and improve modularity of the program.

The control program also included startup delay, motor stopping routines, PWM-based speed control, and settling delays to improve stability during directional changes. The implemented code successfully enabled controlled traversal of the lawn mower robot along the desired path while maintaining stable and reliable operation.

11 Future Scope

Several improvements can be implemented in future versions:

- AI-based navigation
- Computer vision for obstacle detection
- GPS-based path planning
- Automatic grass height adjustment

12 Conclusion

This report presented the complete structural and mechanical analysis of an lawn mower robot using conventional wheeled locomotion.

The chassis was successfully designed and manufactured using 3D printing technology. Detailed analytical calculations and finite element analysis were performed to evaluate structural safety and deformation characteristics.

The ANSYS results confirm that the structure possesses adequate stiffness, low deformation, and a high factor of safety under operational loading conditions.

The developed lawn mower robot provides an efficient and compact solution for grass-cutting applications while demonstrating practical applications of machine design and robotic engineering principles.

References

Shigley, J. E., Mechanical Engineering Design

Norton, R. L., Machine Design: An Integrated Approach

ANSYS Structural Analysis Documentation